

C02 -AT2 -SCENARIO BASED ASSIGNMENT

Q1 AI-Based Student Performance Analytics System

10 marks

AI-Based Student Performance Analytics System Scenario: Intelligent Education. A university wants to develop an AI-assisted Student Performance Analytics System to monitor student academic performance and identify students who may need additional academic support.

- A. Create a class Student with attributes such as Student ID, Name, Marks in different subjects, Average, and Performance Level.
- B. Implement member functions to input student details, calculate average marks, determine performance level, and display the student report.
- C. Use an array of objects to manage records of multiple students.
- D. Include a static member to maintain the total number of students analyzed.
- E. Apply proper encapsulation using private data members and public member functions.

Your Code

```
1 #include <iostream>
2 using namespace std;
3 class Student {
4 private:
5     int id;
6     string name;
7     int marks[3];
8     float average;
9     string level;
10 public:
11     static int total;
12     void input() {
13         cin >> id >> name;
14         for (int i = 0; i < 3; i++)
15             cin >> marks[i];
16         average = (marks[0] + marks[1] + marks[2]) / 3.0;
17         if (average >= 80)
18             level = "Excellent";
19         else if (average >= 60)
20             level = "Good";
21         else if (average >= 40)
22             level = "Average";
23         else
24             level = "Needs Improvement";
25         total++;
26     }
27 }
```

Input

```
101 Pavan
80 90 85
102 Rahul
50 60 55
```

Output



Q2 Digital Healthcare Patient Monitoring System

10 marks

Digital Healthcare Patient Monitoring System ♦ Scenario: Smart Healthcare. A modern hospital wants to implement a Digital Patient Monitoring System that maintains patient information and automatically calculates basic health-risk indicators from recorded medical parameters.

- A. Design a class Patient with attributes such as Patient ID, Name, Age, Heart Rate, Blood Pressure, and Health Status.
- B. Implement member functions to input patient details, calculate/identify the health status based on predefined conditions, and display the patient report.
- C. Store multiple patient records using an array of objects.
- D. Use a static member to track the total number of patients monitored.
- E. Demonstrate data hiding, encapsulation, and structured object-oriented program design.

Your Code

```
1 #include <iostream>
2 using namespace std;
3 class Patient {
4 private:
5     int id, age, heartRate, bp;
6     string name, status;
7 public:
8     static int total;
9     void input() {
10         cin >> id >> name >> age;
11         cin >> heartRate >> bp;
12         if (heartRate >= 60 && heartRate <= 100 && bp >= 90 && bp <= 120)
13             status = "Normal";
14         else
15             status = "Risk";
16         total++;
17     }
18     void display() {
19         cout << "ID: " << id << endl;
20         cout << "Name: " << name << endl;
21         cout << "Age: " << age << endl;
22         cout << "Heart Rate: " << heartRate << endl;
23         cout << "Blood Pressure: " << bp << endl;
24         cout << "Health Status: " << status << endl;
25     }
26 };
27 int Patient::total = 0;
```

Input

```
2
101 Pavan 20
80 110
102 Rahul 25
```

Output



Q3 Digital Wallet and FinTech Transaction System

10 marks

Digital Wallet and FinTech Transaction System ♦ Scenario: Cashless Economy. A FinTech company wants to develop a Digital Wallet Management System to maintain customer wallet information and perform secure financial transactions.

- A. Create a class DigitalWallet with attributes such as Wallet ID, Customer Name, Mobile Number, and Balance.
- B. Implement member functions for wallet creation, money deposit, money withdrawal, balance enquiry, and transaction display.
- C. Use an array of objects to manage multiple customer wallets.
- D. Include a static member to maintain the total number of active wallets.
- E. Apply appropriate access specifiers and encapsulation to prevent direct modification of sensitive financial data.

Your Code

```
1 #include <iostream>
2 using namespace std;
3 class DigitalWallet {
4 private:
5     int id;
6     string name, mobile;
7     float balance;
8 public:
9     static int total;
10    void createWallet() {
11        cin >> id >> name >> mobile >> balance;
12        total++;
13    }
14    void deposit(float amount) {
15        balance = balance + amount;
16    }
17    void withdraw(float amount) {
18        if (amount <= balance)
19            balance = balance - amount;
20        else
21            cout << "Insufficient Balance\n";
22    }
23    void display() {
24        cout << "Wallet ID: " << id << endl;
25        cout << "Customer: " << name << endl;
26        cout << "Mobile: " << mobile << endl;
```

Input

```
2
101 Pavan 9876543210 1000
102 Rahul 9876543211 2000
```

Output



Q4 Smart Agriculture Crop Monitoring System

10 marks

Smart Agriculture Crop Monitoring System ♦ Scenario: IoT-Based Agriculture. An agricultural research centre is developing an IoT-based Smart Agriculture System to monitor crop fields using sensor data and identify whether crops require irrigation.

- Design a class CropField with attributes such as Field ID, Crop Name, Soil Moisture, Temperature, Humidity, and Irrigation Status.
- Implement member functions to input sensor readings, evaluate soil conditions, determine irrigation requirements, and display field information.
- Use an array of objects to maintain information about multiple crop fields.
- Include a static member to count the total number of monitored fields.
- Ensure proper encapsulation and modular object-oriented design suitable for an IoT-based agricultural application.

Your Code

```
1 #include <iostream>
2 using namespace std;
3 class CropField {
4 private:
5     int fieldID;
6     string cropName;
7     float moisture, temperature, humidity;
8     string status;
9 public:
10    static int total;
11    void input() {
12        cin >> fieldID >> cropName;
13        cin >> moisture >> temperature >> humidity;
14        if (moisture < 40)
15            status = "Irrigation Required";
16        else
17            status = "No Irrigation Required";
18        total++;
19    }
20    void display() {
21        cout << "Field ID: " << fieldID << endl;
22        cout << "Crop: " << cropName << endl;
23        cout << "Soil Moisture: " << moisture << endl;
24        cout << "Temperature: " << temperature << endl;
25        cout << "Humidity: " << humidity << endl;
26        cout << "Irrigation Status: " << status << endl;
```

Input

```
101 Rice
30 32 70
102 Wheat
55 28 65
```

Output



Q5 Smart City IoT Energy Management System

10 marks

Smart City IoT Energy Management System ♦ Scenario: Sustainable Smart City. A smart-city research project aims to develop an IoT-based Energy Management System for monitoring electricity consumption in different buildings and identifying buildings with high energy usage.

A. Create a class Building with attributes such as Building ID, Building Name, Building Type, Energy Consumption, and Energy Status.

B. Implement member functions to input building details, calculate energy status based on consumption, and display energy usage information.

C. Use an array of objects to manage multiple buildings in the smart-city system.

D. Include a static member to maintain the total number of buildings monitored.

E. Apply encapsulation, data hiding, and structured C++ object-oriented programming principles in the implementation.

Your Code

```
1 #include <iostream>
2 using namespace std;
3 class Building {
4 private:
5     int id;
6     string name, type, status;
7     float energy;
8 public:
9     static int total;
10    void input() {
11        cin >> id >> name >> type >> energy;
12        if (energy > 500)
13            status = "High";
14        else
15            status = "Normal";
16        total++;
17    }
18    void display() {
19        cout << "ID: " << id << endl;
20        cout << "Name: " << name << endl;
21        cout << "Type: " << type << endl;
22        cout << "Energy: " << energy << endl;
23        cout << "Status: " << status << endl;
24    }
25 };
26 int Building::total = 0;
27 int main() {
```

Input

```
2
101 Office Commercial 650
102 School Educational 300
```

Output